

A Detailed Summary of “Agentic AI in Predictive AIOps: Enhancing IT Autonomy and Performance”

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Abstract—

This report presents the summary of the research article titled “Agentic AI in Predictive AIOps: Enhancing IT Autonomy and Performance” by Shanmugasundaram Sivakumar (2024). The summarized work explores the integration of Agentic Artificial Intelligence within Predictive AIOps platforms and examines how autonomous decision-making, proactive anomaly detection, and self-healing mechanisms enhance IT autonomy and operational performance. The study synthesizes recent literature and case-based evidence to highlight benefits, limitations, and future implications of agentic systems in modern IT infrastructure management.

Keywords: Agentic AI, Predictive AIOps, IT Operations, Anomaly Detection, Resource Optimization, Autonomous Systems

I. INTRODUCTION

The rapid growth of cloud computing, distributed architectures, and large-scale digital services has significantly increased the complexity of IT operations...

II. PREDICTIVE AIOPS

Predictive AIOps utilizes historical and real-time operational data such as logs, metrics, and traces...

III. AGENTIC AI IN IT OPERATIONS

Agentic AI refers to intelligent agents capable of perceiving their environment, making decisions, and executing actions autonomously...

IV. INTEGRATION OF AGENTIC AI WITH PREDICTIVE AIOPS

The integration of Agentic AI with Predictive AIOps creates self-healing and self-managing IT systems...

V. CHALLENGES AND ETHICAL CONSIDERATIONS

Despite its advantages, the deployment of Agentic AI introduces challenges related to explainability, accountability, and data security...

VI. CONCLUSION

The study concludes that Agentic AI significantly enhances the capabilities of Predictive AIOps by enabling autonomous, proactive, and adaptive IT operations.